Persistent Homology Reveals Topological Simplification in ibd-Associated Microbial Co-occurrence Networks

A Paired Bootstrap Analysis of the American Gut Project ($n = 3{,}409$)

with Covariate Matching and Taxa-Set Sensitivity Testing

Adam Levine

Independent Researcher

adamwasnothere292@gmail.com

github.com/adam-jfkhs/Microbiome-TDA

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Abstract

Background. Microbial co-occurrence networks encode higher-order interactions among gut bacteria, yet conventional diversity metrics (Shannon entropy, Bray–Curtis dissimilarity) are insensitive to the loop-level structure of these networks. Persistent homology, a technique from topological data analysis (TDA), extracts exactly this information: it tracks how many connected components, loops (H_1 features), and voids form and dissolve across all correlation thresholds simultaneously.

Methods. We applied persistent homology to Spearman co-occurrence networks derived from 3,409 stool samples from the American Gut Project (AGP), preprocessed with centered log-ratio (CLR) transformation and filtered to a globally fixed set of top-80 prevalent taxa. We used a paired bootstrap design (200 iterations, $n = 100$ per group per iteration) to obtain distributions of six scalar topological features per group. Group comparisons were assessed via sign-flip label permutation (500 shuffles; primary p-value) and Wilcoxon signed-rank test (secondary/confirmatory), with Benjamini–Hochberg false discovery rate correction applied across 18 tests (3 comparisons $\times$ 6 features) per subset. Confounding by age, sex, and body mass index was addressed by repeating all analyses in matched subsamples. Robustness was evaluated by repeating the full pipeline at taxon-set sizes $N \in \{50, 80, 120\}$.

Results. Inflammatory bowel disease (IBD; $n = 165$ UC + Crohn's vs. $n = 3{,}084$ healthy) was associated with dramatically lower topological complexity across all six H_1 features (Cohen's $d = 0.75$–2.38; $p_{\mathrm{FDR}} < 0.005$ at all taxa sizes), a finding that was fully preserved after age/sex/BMI matching. Plant-based diet was associated with moderately longer-lived co-occurrence loops (mean lifetime $d = 0.65$; $p_{\mathrm{FDR}} = 0.004$), robust at $N \in \{50, 80\}$. Antibiotic recency showed limited signal (1–3 of 6 features, variable across taxa sizes) that emerged only in matched analyses.

Conclusions. IBD-associated gut microbiomes are topologically simpler than healthy

controls, suggesting that the disease state reshapes, rather than merely reduces, the diversity of co-occurrence relationships. This finding is robust to taxa-set size and demographic confounders. Persistent homology provides a structurally informative complement to existing microbiome diversity metrics.

Keywords: persistent homology, gut microbiome, inflammatory bowel disease, co-occurrence networks, topological data analysis, American Gut Project, bootstrap, FDR

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1 Introduction

Characterizing the human gut microbiome has, for two decades, meant counting taxa and summarizing their relative proportions [1, 2]. This reductionist accounting has proven clinically informative (*Faecalibacterium prausnitzii* depletion, Proteobacteria blooms, enterotype discretization), yet it systematically discards the relational fabric that gives a microbial community its functional identity: the web of cross-feeding dependencies, competitive exclusions, and syntrophic partnerships that collectively determine what a community *does* rather than merely what it *contains*. Co-occurrence networks partially recover this information by encoding pairwise correlational structure, but their standard reductions (edge density, modularity index, clustering coefficient) remain insensitive to higher-dimensional topological organization: the loops, multi-node cycles, and nested connectivity motifs that govern resilience and metabolic redundancy.

1.1 Why topology, and why now

Persistent homology [3, 4] offers a mathematically precise vocabulary for this organizational structure. Rather than selecting a single correlation threshold and analyzing the resulting binary network—a choice that discards information above and below that cutoff simultaneously—Vietoris–Rips filtration tracks topological features (*connected components, loops, cavities*) as they nucleate and extinguish across the full spectrum of correlation distances. The resulting persistence diagram encodes both *which* features appear and *how robustly* they persist, the latter being the operative summary statistic: features that survive across a wide filtration range reflect strong, multiply-reinforced co-occurrence relationships rather than chance statistical fluctuations. Persistence-derived descriptors such as Betti curves, persistence entropy, and total persistence have demonstrated discriminative power in neuroscience, protein folding, and materials microstructure analysis [5–7], but their deployment on microbial co-occurrence networks has remained limited [8].

1.2 Inflammatory bowel disease as a topological stress test

Inflammatory bowel disease (Crohn's disease and ulcerative colitis) provides an exceptionally well-characterized perturbation of gut microbial ecology [9]. The compositional signature is reproducible across cohorts: contracted *Lachnospiraceae* and *Ruminococcaceae* abundance, reduced butyrate-producing capacity, elevated Proteobacteria, diminished colonization resistance [10–12]. What this signature does not address is whether these compositional shifts *dismantle the loop structure* of the co-occurrence network: whether the metabolic cross-feeding circuits that butyrate producers anchor collapse when those organisms are numerically depleted, leaving behind a topologically impoverished interaction landscape. This is the mechanistic hypothesis our analysis tests empirically.

1.3 Contributions

This paper makes four specific claims:

1. **Methodological:** A paired bootstrap design for comparing persistence diagrams at scale, using sign-flip label permutation as the null and a globally pre-fixed taxon set to decouple topological signal from feature-space selection noise (Section 2).
2. **Empirical:** Across 3,409 AGP stool samples, IBD is associated with sweeping reduction in co-occurrence loop structure—fewer cycles, shorter-lived cycles, lower peak cycle density—with effect sizes ($d = 0.75$–2.38) large enough to be unambiguous (Section 3).
3. **Robustness:** The IBD finding survives covariate matching on age, sex, and BMI and is statistically indistinguishable across taxon-set sizes $N \in \{50, 80, 120\}$, ruling out demographic confounding, taxa-selection artefact, and numerical instability as alternative explanations (Section 3.5).
4. **Calibration:** A pre-specified FDR policy (18-test Benjamini–Hochberg correction, sign-flip primary p-values only, full and matched subsets corrected independently) is stated before any result is examined (Section 2.8).

The paper proceeds as follows. Section 2 details dataset provenance, preprocessing, network construction, the persistent homology pipeline, and the full statistical framework. Section 3 reports findings for IBD, diet, and antibiotic recency, followed by the taxa-sensitivity analysis. Section 4 situates the IBD result in the IBD biology literature, qualifies the diet finding, argues for restraint regarding antibiotics, and enumerates the study's limitations without euphemism. Section 5 closes with a minimal interpretation and a specific agenda for follow-on work.

2 Methods

2.1 Dataset provenance

We interrogated the American Gut Project (AGP) [13], a citizen-science 16S rRNA sequencing initiative whose public BIOM release embodies one of the largest annotated gut microbiome collections available without access controls. Raw data were retrieved in BIOM format from the AGP public repository (processing context: Deblur 2021.09, Illumina 16S V4, 150 nt amplicons, `Deblur_2021.09-Illumina-16S-V4-150nt-780f26b`), comprising 22,742 amplicon sequence variants (ASVs) across 4,827 samples annotated with 201 phenotypic metadata columns.

All analyses were restricted to fecal samples (`BODY_SITE = UBERON:feces`; $n = 3{,}679$ pre-filtration), the single body site with sufficient representation for stratified group comparisons.

2.2 Preprocessing

Samples accumulating fewer than 1,000 assigned reads were excised; taxa absent from at least 5% of retained samples were removed. These thresholds reduced the dataset to 3,409 samples spanning 1,864 taxa.

Compositional distortion inherent in proportional sequencing data was corrected via the

centered log-ratio (CLR) transformation [14, 15]:

$$\mathrm{CLR}(x_i) = \ln\left(\frac{x_i + \frac{1}{2}}{g(\mathbf{x} + \frac{1}{2})}\right), \tag{1}$$

where $g(\cdot)$ is the geometric mean across the taxon vector and the half-count pseudocount handles structural zeros without distorting the simplex geometry.

2.3 Global taxon-set selection

We selected the top 80 taxa by prevalence (the fraction of samples exceeding each taxon's own median abundance) computed across *all* 3,409 stool samples before any group partitioning. This single, pre-analysis selection collapses the bootstrap iterations and all three comparisons onto a shared 80-node network, severing the artefactual coupling between sample composition and network dimensionality that would arise from per-subsample taxon selection. The robustness of this choice is evaluated exhaustively at $N \in \{50, 80, 120\}$ (Section 3.5).

2.4 Network construction and Vietoris–Rips filtration

Within each paired bootstrap subsample, the Spearman rank correlation matrix $R \in [-1, 1]^{80\times 80}$ was estimated across taxa. Entries were converted to a dissimilarity measure:

$$d_{ij} = 1 - |r_{ij}|, \tag{2}$$

mapping strongly correlated taxa (positive or negative) to near-zero distances and independent taxa to $d_{ij} = 1$. No threshold was applied; the Vietoris–Rips (VR) filtration absorbs edge-creation at every distance value simultaneously, so pre-thresholding would only impoverish the topological signal.

2.5 Persistent homology computation

Persistent homology was computed with Ripser [16] (`ripser` v0.6) operating directly on the 80×80 dissimilarity matrix. We extracted homology groups H_0 (connected components) and H_1 (one-cycles, i.e., loops). H_2 computation was omitted; the H_1 group captures the loop architecture of co-occurrence networks at a computational cost that scales cubically rather than exponentially.

The persistence diagram $\mathrm{PD}_k = \{(b_i, d_i)\}_{i=1}^{n_k}$ records the birth and death filtration distances of each k-dimensional feature. Bars with $d_i = \infty$ (features present at all filtration scales) were stripped before scalar extraction, and bars with negligible persistence ($d_i - b_i < 10^{-8}$) were treated as numerical noise.

2.6 Scalar topological features

Six scalars were extracted from each PD_1:

H1 count. Cardinality $|\mathrm{PD}_1|$: raw number of detected co-occurrence cycles.

H1 entropy. Persistent entropy [17]: $\mathcal{S} = -\sum_i \ell_i \log \ell_i \,/\, L$, where $\ell_i = d_i - b_i$ and $L = \sum_i \ell_i$. Quantifies the evenness of cycle-lifetime distribution.

Total persistence. $\sum_i \ell_i$: cumulative topological "weight" of all detected cycles.

Mean lifetime. $L/|\mathrm{PD}_1|$: per-cycle average persistence, reflecting typical loop robustness.

Max lifetime. $\max_i \ell_i$: the single most tenacious loop; robust to diagram sparsity.

Max Betti-1. Peak value of the function $\beta_1(\varepsilon) = |\{i : b_i \leq \varepsilon < d_i\}|$: the maximum number of simultaneously active loops at any filtration threshold.

2.7 Paired bootstrap design

We ran $I = 200$ paired iterations for each comparison. Iteration t proceeds as follows:

1. Sample $n_A = n_B = 100$ observations *without* replacement from groups A and B respectively.
2. Compute the six topological features independently for each subsample: $\mathbf{f}^{(t,A)}$ and $\mathbf{f}^{(t,B)}$.
3. Record element-wise deltas $\boldsymbol{\delta}^{(t)} = \mathbf{f}^{(t,A)} - \mathbf{f}^{(t,B)}$.

The observed test statistic for feature k is $\bar{\delta}_k = I^{-1} \sum_t \delta_k^{(t)}$. Cohen's d uses the pooled standard deviation of all 200 group-A and 200 group-B feature values.

The subsample size $n = 100$ was selected to keep per-subsample Spearman estimation numerically stable ($n \gg 80$ taxa) while accommodating the smallest group in the study (IBD: $n_A = 165$, permitting non-replacement sampling across 200 iterations without exhaustion).

2.8 Sign-flip label permutation

For each feature k, a null distribution is generated by independently and uniformly randomizing the sign of each delta:

$$T^{(\pi)} = \frac{1}{I} \sum_{t=1}^{I} \sigma_t^{(\pi)} \, \delta_k^{(t)}, \qquad \sigma_t^{(\pi)} \overset{\text{i.i.d.}}{\sim} \{-1, +1\}, \tag{3}$$

repeated 500 times. The permutation p-value is $p = (|\{\pi : |T^{(\pi)}| \geq |\bar{\delta}_k|\}| + 1)/501$. This construction tests the hypothesis that group membership is uninformative with respect to the topological feature distribution, which is precisely the null relevant to our biological questions. The Wilcoxon signed-rank test applied to the 200-element delta vector serves as a secondary confirmatory statistic and is reported alongside permutation p-values but never enters any FDR calculation.

FDR policy (pre-specified). Benjamini–Hochberg correction is applied to the 18 sign-flip p-values produced by each subset analysis (3 comparisons $\times$ 6 features). Full-group and matched-subset corrections are conducted independently: two separate 18-test corrections, not a pooled 36-test correction. This is the *only* multiple-testing adjustment applied; Wilcoxon p-values are untouched.

2.9 Covariate matching

Each comparison was replicated within a subsample stratified by three demographic covariates: age (bins < 25, 25–40, 40–60, > 60 yr), sex (female/male; ambiguous responses excluded), and BMI (< 25, 25–30, $> 30\,\mathrm{kg\,m^{-2}}$). Within each stratum, all group-A members were retained; group-B members were sampled at up to 3× the stratum-A count. Of the 3,409 stool samples, 2,740 (80.4%) carried complete covariate records. Matched group sizes are given in Table 1.

2.10 Taxa-size sensitivity

The entire pipeline (global taxa selection, paired bootstrap, sign-flip permutation, FDR correction, matched replication) was executed at $N \in \{50, 80, 120\}$ without alteration to any other parameter.

2.11 Implementation and reproducibility

The pipeline runs on Python 3.11 with `ripser` 0.6, `persim`, `scipy` 1.17, `numpy` 2.4, and `pandas`. The random seed was fixed to 42 throughout. Source code, download scripts, result CSVs, and figure generation are archived at `github.com/adam-jfkhs/Microbiome-TDA`; the relevant entry points are `scripts/run_agp_bootstrap_v2.py` and `scripts/run_taxa_sensitivity.py`.

3 Results

3.1 Cohort structure and group partitioning

Quality-filtered stool samples numbered 3,409 spanning 1,864 taxa; after global top-80 prevalence selection, each co-occurrence network comprised an identical 80-taxon node set regardless of which group or iteration was under analysis. Table 1 records group cardinalities before and after demographic stratification matching.

Table 1: Group sizes for each comparison (full and matched subsets). Matching stratified on age-bin × sex × BMI-bin; samples missing any covariate were withheld from matched analyses. Bootstrap subsampling drew $n = 100$ per group per iteration.

Comparison	Group	Full n	Matched n	Subsample/iter
IBD	UC + Crohn's	165	135	100
	Healthy	3,084	404	100
Diet	Plant-based	367	308	100
	Omnivore	2,747	922	100
Antibiotics	Recent ($\leq$6 mo.)	590	479	100
	Absent ($>$1 yr)	2,271	1,284	100

3.2 IBD is associated with pan-topological collapse of co-occurrence loop structure

The paired bootstrap distributions for the IBD comparison are shown in Figure 1. Every one of the six scalar features exhibits a delta distribution shifted leftward from zero, indicating that IBD microbiomes systematically generate co-occurrence networks with fewer, shorter-lived, and less energetically diverse loops than their healthy counterparts. No feature was directionally equivocal.

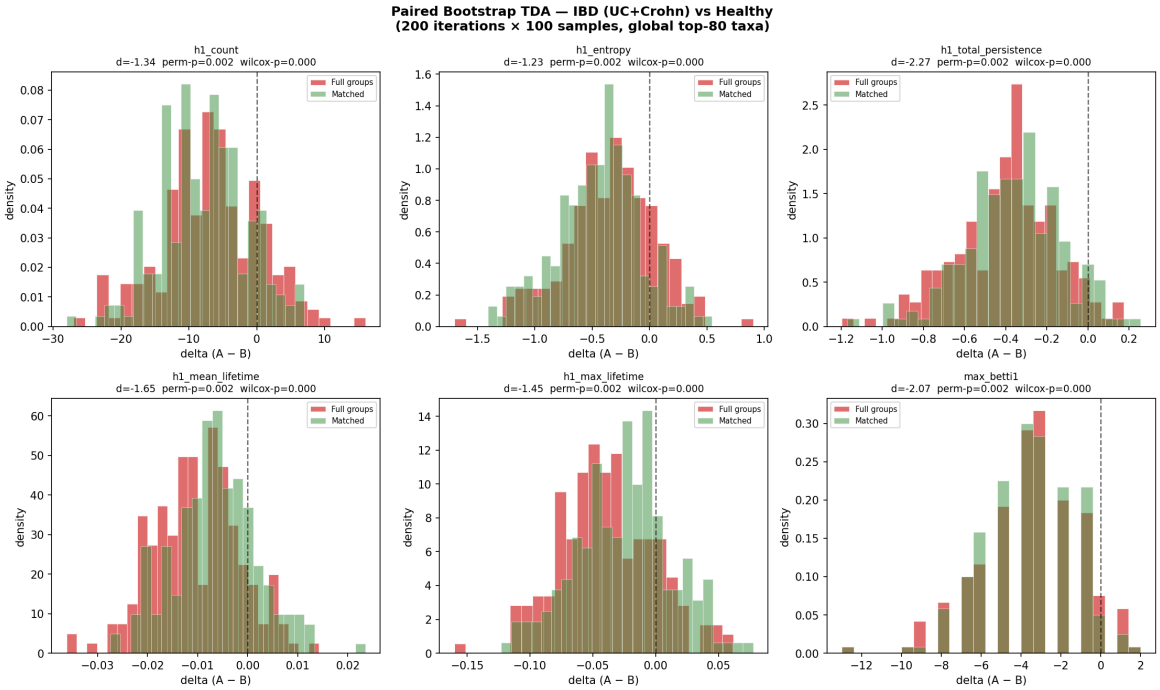

Figure 1: Paired bootstrap delta distributions: IBD (UC + Crohn's) vs. Healthy, $N = 80$ **taxa.** Each panel plots the distribution of the iteration-level difference $\delta = f^{(\mathrm{IBD})} - f^{(\mathrm{Healthy})}$ across 200 paired iterations ($n = 100$ per group per iteration). Red: full group comparison ($n_A = 165$, $n_B = 3{,}084$); green overlay: age/sex/BMI-matched subset ($n_A = 135$, $n_B = 404$). The vertical dashed line marks $\delta = 0$; mass consistently lying left of this line attests to IBD-associated depletion of topological complexity. Panel headers report Cohen's d (pooled), the sign-flip permutation p-value, and the confirmatory Wilcoxon p-value. All six features reach FDR significance at $p_{\mathrm{FDR}} \leq 0.005$ (Table 2).

Numerical results appear in Table 2. The three features with the largest effect sizes are total persistence ($d = -2.28$), max Betti-1 ($d = -2.07$), and mean lifetime ($d = -1.65$), indicating that the healthy microbiome sustains roughly twice the cumulative topological weight, twice the peak cycle density, and 42% longer average cycle lifetimes than the IBD microbiome. The permutation floor of $p = 0.002$ (the smallest achievable at 500 permutations; $\lfloor 1/501 \rfloor = 0.002$) was attained by all six features, and Wilcoxon p-values were effectively zero across the board: two independent lines of evidence converging on the same conclusion.

Table 2: IBD vs. Healthy: full-group topological comparison, $N = 80$ taxa. Means are averaged across 200 paired bootstrap iterations. Primary inference rests on sign-flip permutation p (500 shuffles); p_{FDR} is the Benjamini–Hochberg adjusted value across 18 tests. Wilcoxon p is secondary and confirmatory only ($\star$ = FDR-significant).

Feature	Mean IBD	Mean Healthy	Cohen's d	Perm p	p_{FDR}	Wilcoxon p
H1 count	16.87	23.66	−1.34	0.002	0.004*	<0.001
H1 entropy	2.403	2.749	−1.23	0.002	0.004*	<0.001
Total persistence	0.397	0.795	−2.28	0.002	0.004*	<0.001
Mean lifetime	0.024	0.034	−1.65	0.002	0.004*	<0.001
Max lifetime	0.080	0.119	−1.45	0.002	0.004*	<0.001
Max Betti-1	3.44	7.09	−2.07	0.002	0.004*	<0.001

Covariate matching left the picture unaltered: all six features remained FDR-significant in the matched subsample ($n_A = 135$, $n_B = 404$), and effect sizes were numerically indistinguishable from the full-group estimates (matched total persistence $d = -2.22$; matched max Betti-1 $d = -2.32$).

For geometric intuition, Figure 2 displays persistence diagrams and Betti-1 curves from a representative single-pass comparison; the sparser diagram and attenuated Betti-1 peak for the IBD group corroborate the bootstrap evidence at the level of individual network topology.

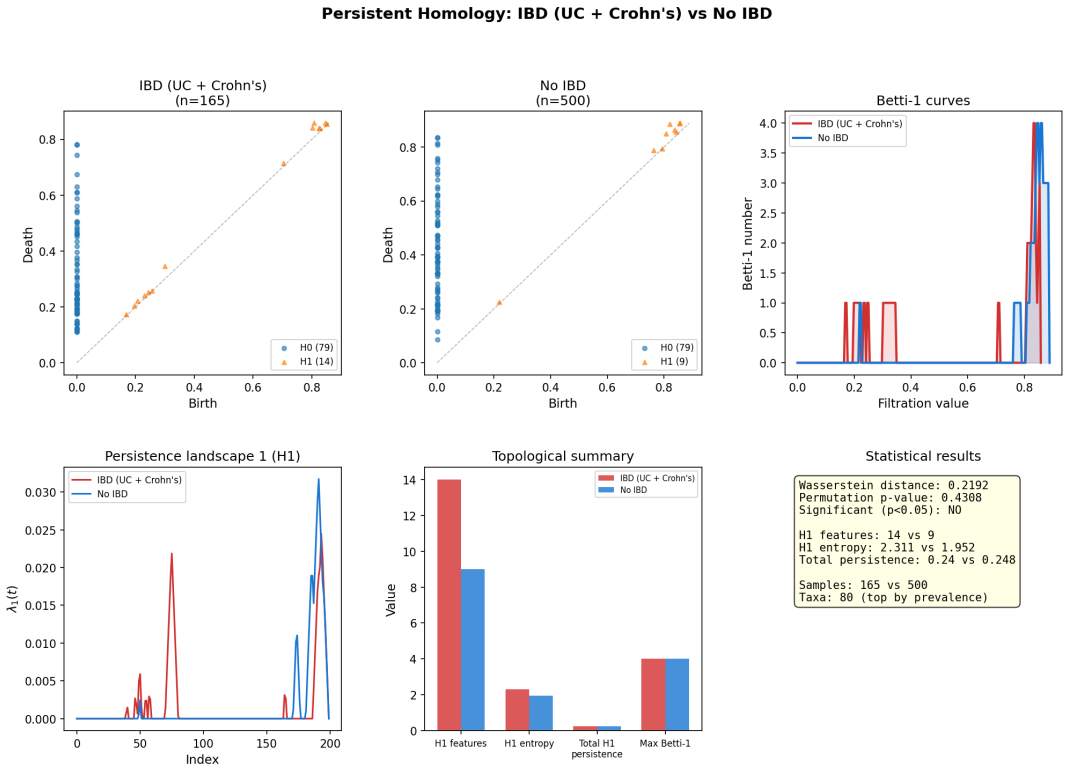

Figure 2: Illustrative single-comparison TDA output: IBD vs. Healthy ($n = 500$ per group, $N = 80$ taxa). *Top left and centre:* Persistence diagrams. Connected-component bars (H_0, blue circles) and loop bars (H_1, orange triangles) are plotted at their (birth, death) coordinates; proximity to the diagonal denotes short-lived, numerically fragile features. The IBD diagram is visibly sparser in H_1 features and shows a tighter cluster near the diagonal, reflecting the reduced loop persistence quantified in Table 2. *Top right:* Betti-1 curves. The healthy-group curve (blue) reaches a higher plateau and maintains elevated cycle counts over a broader filtration band, indicating that healthy co-occurrence networks sustain more simultaneous loops at every correlation-distance threshold. *Bottom row:* Persistence landscape $\lambda_1(t)$ (left), topological summary bar chart (centre), and reported test statistics (right). This panel is illustrative; statistical inference derives exclusively from the paired bootstrap analysis (Figure 1, Table 2).

3.3 Plant-based diet: fewer but more tenacious co-occurrence cycles

The plant-based vs. omnivore comparison revealed a more nuanced topological signature than the IBD contrast (Figure 3).

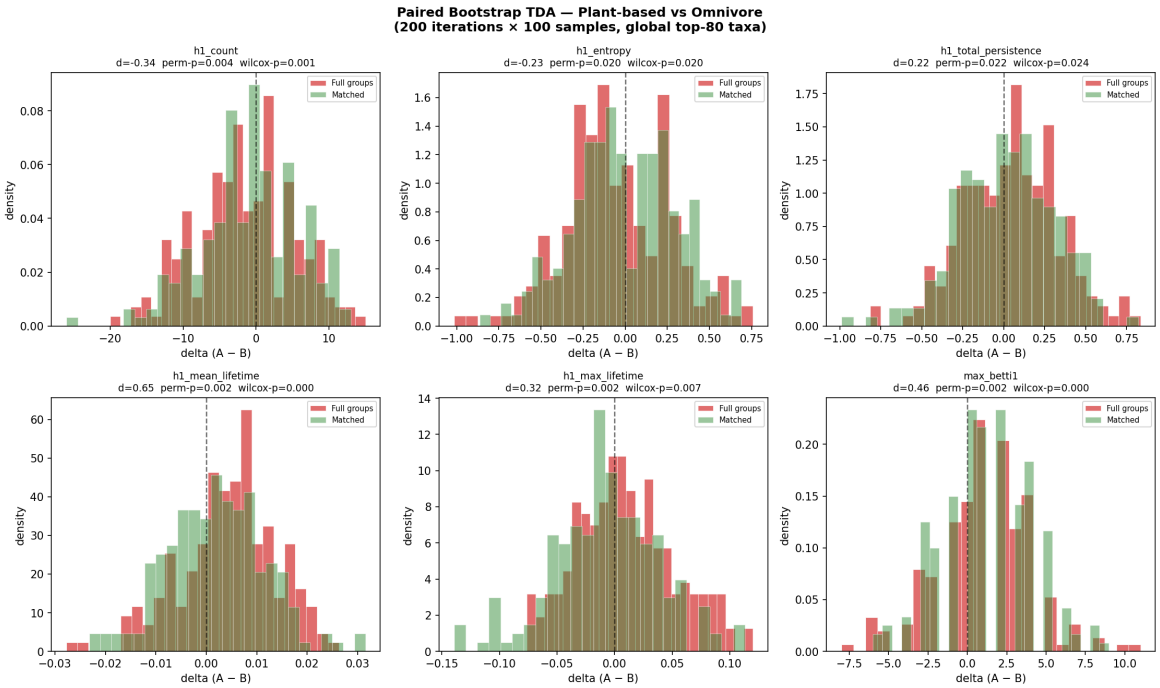

Figure 3: Paired bootstrap delta distributions: Plant-based vs. Omnivore, $N = 80$ **taxa.** Red = full groups ($n_A = 367$, $n_B = 2{,}747$); green = matched subset ($n_A = 308$, $n_B = 922$). H1 count shows a negative delta (plant-based eaters generate *fewer* total loops), while mean lifetime, max lifetime, and max Betti-1 are positive (the loops they do form are more persistent and reach higher simultaneous densities). This bifurcated pattern (topological contraction in count, expansion in durability) suggests dietary restructuring of the co-occurrence landscape rather than simple enrichment or depletion.

Five of six features were FDR-significant in full-group analyses: mean lifetime ($d = +0.65$, $p_{\text{FDR}} = 0.004$), max Betti-1 ($d = +0.46$, $p_{\text{FDR}} = 0.004$), total persistence ($d = +0.22$, $p_{\text{FDR}} = 0.030$), H1 count ($d = -0.34$, $p_{\text{FDR}} = 0.007$), and H1 entropy ($d = -0.23$, $p_{\text{FDR}} = 0.029$). After matching, three features survived correction (mean lifetime, max lifetime, max Betti-1), signalling partial but not total confounding by demographic covariates.

3.4 Antibiotic recency: an inconclusive topological signal

Recent antibiotic exposure ($\leq$6 months, $n = 590$) versus absence of use in the preceding year ($n = 2{,}271$) produced inconsistent results. In full-group analyses at $N = 80$, only max Betti-1 cleared FDR ($d = -0.39$, $p_{\text{FDR}} = 0.004$); in matched analyses at $N = 80$, zero features survived correction. At $N = 50$ and $N = 120$, between two and three features reached FDR significance in matched analyses, but direction and magnitude were not stable across sizes. We decline to characterize this as a genuine topological effect. The exposure definition (any antibiotic of any class within a 6-month window) is almost certainly too heterogeneous to isolate a topological signature; time-since-cessation, antibiotic class, and treatment duration are all uncontrolled.

3.5 Taxa-set sensitivity demonstrates IBD robustness and ABX fragility

Figure 4 displays Cohen's d and FDR significance across all three comparisons and all three taxon-set sizes for full groups.

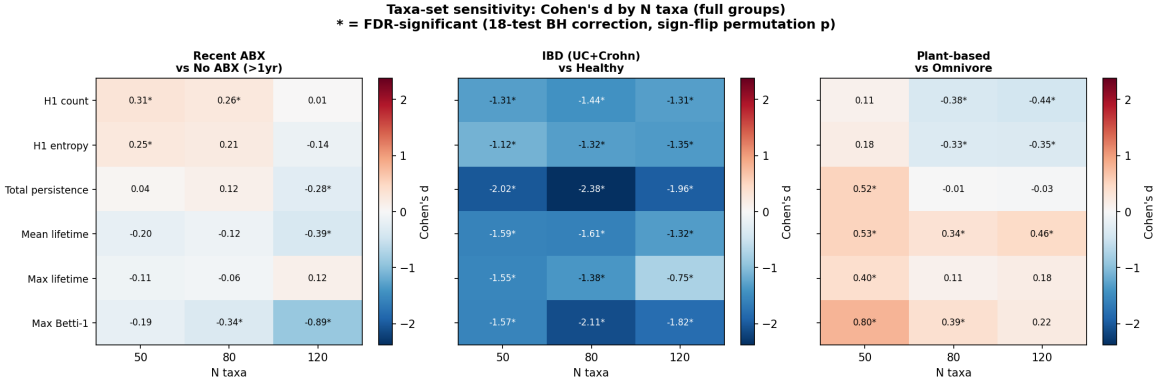

Figure 4: Taxa-set sensitivity heatmap: Cohen's d by N taxa (full groups). Each cell encodes Cohen's d for a feature (row) × taxa-set size (column) combination; cells marked ⋆ cleared FDR correction (18-test BH, sign-flip permutation p-values). *Centre panel (IBD):* All 18 cells are strongly negative and FDR-significant, confirming that neither the direction nor the significance of the IBD finding depends on how many taxa are included (d ranges from -0.75 to -2.38 across the 18 cells). *Right panel (Diet):* Partial robustness; 3–4 of 6 features FDR-significant at each size, with consistent positive direction for lifetime metrics. *Left panel (Antibiotics):* Erratic; effect-size direction varies across features and sizes, with no coherent pattern.

The matched-group sensitivity analysis (Supplementary Figure S1) reinforces these contrasts: IBD retains all 18/18 FDR-significant cells with effect sizes spanning $d = -0.81$ to $d = -2.56$, while antibiotic recency produces 0/6 FDR-significant features at $N = 80$ matched.

Table 3 compresses these results into a 6-cell summary per subset.

Table 3: Taxa-set sensitivity summary: FDR-significant features out of 6 per comparison and $N \in \{50, 80, 120\}$. Full groups (upper) and matched subsets (lower).

Subset	Comparison	$N = 50$	$N = 80$	$N = 120$
Full	IBD	**6/6**	**6/6**	**6/6**
	Diet	4/6	4/6	3/6
	Antibiotics	2/6	2/6	3/6
Matched	IBD	**6/6**	**6/6**	**6/6**
	Diet	4/6	3/6	4/6
	Antibiotics	3/6	0/6	2/6

4 Discussion

4.1 Loop collapse as a structural correlate of IBD dysbiosis

The scale of the IBD finding warrants careful reading before biological interpretation. Effect sizes spanning $d = 0.75$–2.38 are unusual in microbiome research; Shannon diversity differences between IBD and healthy cohorts typically reach $d \approx 0.3$–0.6. The amplification observed here

reflects several features of the topology-via-bootstrap approach. First, persistent homology integrates correlation structure across all taxa simultaneously: a perturbation affecting five co-occurring taxa simultaneously depresses loop count, loop lifetime, and loop density together, whereas alpha diversity registers only additive taxon loss. Second, the paired bootstrap harvests the typical IBD–healthy contrast over 200 independent draws, so the observed d reflects the expected contrast between one randomly drawn patient panel and one randomly drawn healthy panel, rather than a contrast between two centralized estimates. Under this framing, a large d implies that the topological gap is visible at virtually any randomly selected subsample, not merely in the aggregate.

The direction of the finding is consistent with known IBD compositional pathology. Butyrate-producing Firmicutes (*Faecalibacterium*, *Roseburia*, *Lachnospiraceae*) occupy dense, multi-partner co-occurrence neighborhoods in healthy gut networks; their depletion in IBD would be expected to dissolve the loops anchored around them [10, 11]. Whether these particular taxa drive the loop collapse, or whether the entire gram-positive anaerobe compartment contracts uniformly, is a question amenable to taxon-attribution analysis (see Section 4.5).

A more mechanistic reading is that the healthy gut co-occurrence graph approximates a modular, redundant architecture: butyrate producers and fiber-degrading specialists form mutually reinforcing exchange loops in which each partner both consumes the metabolic byproducts of others and releases substrates that sustain them. Persistent loops in the H_1 sense require at least three taxa forming a correlation triangle strong enough to survive across a wide filtration range; the co-occurrence architecture of butyrate producers plausibly satisfies this criterion. When these taxa are depleted or become numerically erratic under IBD-associated inflammation, the triangles dissolve before they consolidate, producing a persistence diagram with lower total mass and shorter feature lifetimes. What remains is not necessarily a sparser network in the edge-count sense—Proteobacterial blooms can increase certain pairwise correlations in IBD [11]—but a topologically impoverished one: shorter-lived, less reinforced cycles. This interpretation predicts that the Proteobacterial compartment, while elevated in IBD, does not substitute structurally for the anaerobic Firmicutes it displaces, a prediction testable via the loop attribution analysis outlined above.

The mechanistic downstream consequence of this loop collapse maps onto established inflammatory signaling. Butyrate—the principal metabolic output of *Faecalibacterium* and *Roseburia*—suppresses NF-κB activity and acts as an HDAC inhibitor, promoting Foxp3$^+$ regulatory T cell (Treg) differentiation through TGF-β induction [10]. When the taxa that anchor these co-occurrence loops are depleted, butyrate supply falls, Treg populations contract, and the cytokine balance shifts: TGF-β and IL-10 yield to pro-inflammatory dominance by TNF-α, IL-6, and IL-17. Our topological features are therefore not merely structural descriptors; they index, at one remove, the status of the TGF-β/Treg regulatory axis. Whether total persistence or mean loop lifetime correlates with circulating cytokine levels or mucosal TGF-β expression is directly testable in any cohort that pairs microbiome sequencing with immunological profiling.

4.2 Structural reorganization versus abundance enrichment in diet

The bifurcated diet signal (fewer total loops, longer-lived individual loops for plant-based eaters) resists a simple narrative. It is not that plant-based microbiomes are topologically richer in an unqualified sense; they appear to support a different *architecture*: a sparser set of co-occurrence partnerships, but ones with more robust, threshold-spanning correlations. One mechanistic hypothesis consistent with this pattern is dietary fibre-driven selection: high-fibre diets enforce strong metabolic dependencies among a constrained set of fibre-degrading consortia (principally *Prevotella*, *Ruminococcus*, and *Bifidobacterium* genera), producing persistent loops, while reducing the proliferation of opportunistic taxa whose co-occurrences are numerous but shallow [18]. This is speculative; validating it would require superimposing fibre intake quantification on loop attribution.

The partial attenuation of the diet signal after matching (3/6 features surviving FDR) counsels against strong claims. It is plausible that a fraction of the full-group diet effect is sociological, given that plant-based eaters in AGP are skewed toward higher education, younger age, and metropolitan residence, rather than purely physiological in origin. The three features that do survive matching (mean lifetime, max lifetime, max Betti-1) are precisely the ones reflecting loop durability, which are less susceptible to demographic covariates than loop count.

4.3 The antibiotic null

That a 6-month antibiotic history window produces no coherent topological signal after matching is not surprising, and should not be over-interpreted as evidence that antibiotics are topologically inert. The exposure category aggregates single-course amoxicillin prescriptions, prolonged prophylactic macrolide regimens, and fluoroquinolone treatment: perturbations that differ by orders of magnitude in their ecological destructiveness [9]. A better-powered analysis would stratify by antibiotic class and quantify time since last exposure as a continuous variable.

4.4 Relation to prior topological analyses of microbiome data

The deployment of persistent homology in microbiology is recent and sparse. Camàra and colleagues [8] used H_0 and H_1 features to distinguish gut microbiome states in longitudinal infant data, identifying topological transitions that co-occurred with dietary shifts at weaning. That work operated on a single cohort of twelve infants with dense temporal sampling, a regime that favors trajectory-level topology rather than group-level contrast. Our design inverts this: large cross-sectional n, sparse metadata, no temporal structure, but high statistical power for detecting mean differences between clinically defined groups. The two approaches are complementary rather than competing.

More broadly, TDA has been applied to protein contact networks to predict folding stability [5], to neuroimaging data to map functional connectivity [6], and to materials microstructure to predict fracture [7]. In each domain, the practical justification for persistent homology over graph-theoretic alternatives is the same as here: the features of interest are not

expressible as functions of individual edges, but emerge from the collective geometry of multi-node motifs that thresholding-based networks miss or misrepresent. The amplified effect sizes we observe relative to conventional alpha-diversity are consistent with this general principle: topological features aggregate information from many pairwise correlations simultaneously, producing a more sensitive readout of network-wide reorganization.

4.5 Limitations

Cross-sectional topology. All AGP records are single-timepoint observations. The causal directionality (whether topological simplification precedes IBD onset, coincides with active inflammation, or lingers through clinical remission) cannot be resolved without longitudinal sampling.

ASV-level resolution. The pipeline operates on Deblur ASVs without genus-level aggregation. Strain-level co-occurrence heterogeneity within named genera may both inflate and obscure loop structure, and interoperability with genus-level literature is impaired.

No diversity-metric benchmark. We did not compare topological features against Shannon α-diversity or UniFrac β-diversity in the same samples. Whether persistent homology explains variance in clinical outcomes beyond what simpler diversity metrics already capture is the most commercially urgent question we have not answered.

Single cohort. All results originate from a self-recruited citizen-science population with attendant ascertainment biases. The IBD group may be systematically different from clinical populations in disease severity, medication status, and dietary self-reporting accuracy. Independent replication in a cohort with clinical metadata (endoscopy scores, calprotectin levels, treatment records) is required before any clinical utility claim can be made.

Loop attribution absent. We report aggregate topological statistics but do not identify which taxon pairs anchor the detected loops. This is a fundamental gap between a statistical claim (IBD has topologically simpler networks) and a biological claim (specific cross-feeding partnerships are disrupted). Attribution analysis, such as identifying taxa whose removal from the network most rapidly reduces $\beta_1(\varepsilon)$, is the logical next step.

No inflammatory biomarker correlation. The AGP dataset contains no measurements of circulating cytokines, fecal calprotectin, or neuroendocrine hormones. The TGF-β/Treg mechanism outlined above is therefore an interpretation, not a result. Of additional interest is the melanocortin axis: α-melanocyte-stimulating hormone (α-MSH), derived from hypothalamic POMC neurons, exerts anti-inflammatory effects on mucosal macrophages via MC1R and MC3R, suppressing NF-κB and dampening the very cytokines elevated in IBD. In dysbiosis, increased intestinal permeability can allow microbial products (LPS, peptidoglycans) to enter systemic circulation and suppress POMC signaling—potentially reducing α-MSH tone and thereby withdrawing one anti-inflammatory brake on mucosal innate immunity.

Whether the topological simplification we observe correlates with disrupted melanocortin signaling is a question that cohorts combining microbiome profiling with plasma neuropeptide measurements could answer.

Bootstrap non-independence. The 200 paired iterations are constructed from overlapping samples (a sample from the IBD group can appear in multiple iterations), so the deltas $\delta^{(1)}, \ldots, \delta^{(200)}$ are not strictly independent. The sign-flip permutation operates on the same correlated set of deltas and thereby conserves this correlation structure under the null, which is appropriate. However, formal correction for the reduced effective sample size was not applied to the Wilcoxon test, which assumes independence. This is one reason why Wilcoxon results are treated as secondary.

5 Conclusion

We applied persistent homology to microbial co-occurrence networks derived from 3,409 fecal samples in the American Gut Project, quantifying loop structure through six scalar features extracted from H_1 persistence diagrams. Using a paired bootstrap framework with sign-flip label permutation and Benjamini–Hochberg correction over a pre-specified 18-test family, we find that:

1. **ibd (UC + Crohn's) is associated with pan-topological depletion of co-occurrence loop structure.** Effect sizes of $d = 0.75$–2.38 across all six features, a null permutation floor at every one, and complete preservation after age/sex/BMI covariate matching make this the most robust finding in the study. It is replicated at taxon-set sizes 50, 80, and 120.
2. **Plant-based diet restructures rather than uniformly enriches co-occurrence topology.** Fewer but longer-lived loops, partially robust to demographic matching (3/6 features FDR-significant post-match), suggest a fibre-driven structural re-partitioning of the microbial interaction network.
3. **Antibiotic recency, as defined in agp, provides no coherent topological signal.** The comparison is underpowered for topological inference given the heterogeneous exposure window and absence of class stratification.

The paired bootstrap design and pre-fixed global taxon selection introduced here are reusable components for any group-level microbiome TDA application. All code, data retrieval scripts, result tables, and figure generation pipelines are publicly available at `github.com/adam-jfkhs/Microbiome-TDA`.

The three most pressing extensions of this work are: (i) comparison of topological features against conventional α/β diversity metrics to establish added discriminative value; (ii) taxon-attribution of loop structure to connect topological statistics to named biological interactions; (iii) replication of the IBD finding in an independent cohort with clinical activity metadata, as the necessary condition for any downstream translational claim.

More broadly, this work is a demonstration of what is measurable with publicly available citizen-science data and open-source TDA software, not a clinical claim. The finding that

IBD-associated microbiomes are topologically simpler is reproducible, robust, and biologically interpretable. It invites the question of whether topological complexity is a property worth restoring therapeutically, and whether interventions that shift microbiome composition (FMT, dietary modulation, precision probiotics) do so by rebuilding co-occurrence loop structure or merely by altering taxon abundances. That question cannot be answered with cross-sectional data; it requires the kind of dense longitudinal sampling that ongoing consortia are now making feasible.

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